Using parametric effectiveness for efficient CAD-based automotive design driven by adjoint sensitivity maps

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Summary

This paper demonstrates an efficient CAD-Based adjoint optimization for flow problems, which exploits a concept called parametric effectiveness to select the optimum sets of parameters to use in optimization. Parametric Effectiveness is computed as the ratio of change in performance achieved by perturbing the parameters in the optimum way, to the change in performance that would be achieved by allowing the boundary of the model to move without the constraint on shape change enforced by the CAD parameterization for the same overall boundary movement. An automated process based on a power-set approach is developed which uses parametric effectiveness of individual CAD parameters to select the combination of parameters which would provide the greatest performance improvement. The approach is applied to the shape optimization of an automotive car mirror for noise reduction, which targets the minimization of a surrogate objective function related to aeroacoustics. The flow sensitivitivies are computed with the continuous adjoint method, formulated to include the fully differentiated Spalart-Allmaras turbulence model based on wall functions.

Keywords*: parametric effectiveness, continuous adjoint method, design velocity, surrogate aeroacoustics, adjoint turbulence model*

1 Introduction

Industrial component design typically involves the creation of a CAD geometry which is used as the basis for optimization, and after optimization the design has to be available as an updated CAD model which could be used for optimization. However, currently there is no practicable robust way of either optimising CAD geometries directly or generating a new or updated CAD model from the optimisation results. Where the CAD model was not optimized directly, authors have attempted to optimise the model shape by modifying the computational mesh used for the flow analysis. The main drawback of these approaches is that the mesh topology must remain constant as the model updates, and it is the mesh that represents the optimum shape[1]. This leads inevitably to difficulties integrating the resulting shape back into the CAD-based design workflows.

When optimizing the CAD geometry directly one of the methods is to use a NURBS geometry generated from CAD modelling software, where the position of the NURBS control points are the optimisation variables[2]. The main drawback of this approach is that while the geometry is available as a CAD model, all of the information about which features were used to build the model are lost. In many cases the design intent for the model is inherent in these features. Another option is to use the feature based CAD model with all of the construction history to preserve the design intent. In this approach using the CAD feature parametric information, the shape of a part model can be updated by changing the values of the parameters which define it. The main advantage of this approach is that as the optimized model exists in the CAD system it can be directly used for downstream applications, including manufacturing and process planning. The downside of this approach is that often it is not obvious from the parameterisation which parameter value(s) need to be modified to achieve the desired shape change, especially

when the person implementing the change is not the creator of the CAD model. In industry today, the resulting modification becomes a case of trial and error, which is time consuming for simple models with tens of parameters, and is unfeasible for complex component models with hundreds or thousands of parameters. It is also difficult to maintain a rationale for perturbing multiple parameters simultaneously.

A promising method for efficient CFD optimisation is the adjoint approach which allows the computation of gradients at cost that is independent of the number of design variables, and has been extensively developed and used in the last two decades[3-5]. This is achieved by additionaly solving the adjoint problem at a computational cost comparable to that of the primal analysis. To derive the adjoint PDEs and their boundary conditions the objective function is firstly augmented with the volume integrals of the flow equations residuals multiplied with the corresponding adjoint variables. Afterwards, the variation of the augmented function is processed, so as to eliminate the terms expressing the variation of primal variables with respect to the design variables. The adjoint equations are then discretized and solved to compute the sensitivity derivatives.

Some authors[6,7] attempted to link the adjoint sensitivity derivatives directly with CAD models, so the resulting design is available as a CAD model. Robinson *et. al.*[8] used adjoint methods to define parametric effectiveness which rates the quality of parameters in the CAD model for use as optimization variables. This paper contributes to this concept, by using a power-set approach to select a subset of parameters which provides the greatest potential for performance improvement. While one of the benefits of adjoint optimization is that the cost of calculating sensitivities is virtually independent of the number of design variables, modifying the CAD model by using all parameters in the optimization step is potentially costly, and so identifying a key set from all parameters is advantageous.

In this paper, the developed framework is applied to an automotive noise reduction problem, with the use of a surrogate model for aeroacoustics[9]. Moreover, the continuous adjoint method with the fully differentiated Spalart-Allmaras turbulence model based on wall functions, as developed by Papoutsis *et. al.*[10], is applied. The model under investigation is the TUM DrivAer vehicle[11], using a fast-back configuration with smooth underbody and closed wheels. The optimization procedure alters the shape of the mirror geometry, targeting at a shape which transmits less noise to the interior. The flow equations are solved using the standard steady state incompressible OpenFOAM© solver simpleFoam. The adjoint equations are solved using the Helyx® Adjoint solver, provided by ENGYS[12].

The remainder of the paper will first summarize the CAD modelling approach, followed by the continuous adjoint formulation for surrogate aeroacoustics. This will be followed by a methodology to select the efficient CAD parameters and results obtained for the automobile test case. The paper will finish with the conclusions.

2 CAD feature modelling and Design velocity computation

In a feature based CAD modelling system like CATIA V5 which is a popular industrial CAD system, a part model is comprised of individual features which are combined together to represent an overall shape. Examples of CAD features include pads, pockets, holes, fillets, chamfers etc. In CATIA V5 these features are classified as either sketch-based features or dress-up features. Sketch-based features are created by defining a 2D sketch profile for which a 3D feature is generated from the sketch using operations such as extrusion, rotation, sweeping or lofting. Dress-up features, like fillets and chamfers, are created directly on the solid model.

In this work the CAD model of the car mirror was originally provided as a STEP file. This is a CAD translation standard that does not include any features or parameters. A replica model was created using CATIA V5 using a series of points, lines and splines. The wireframe of the mirror CAD model is shown in Figure 1. The surface fitting methods in CATIA V5 (like multi-section surface and fill surface) were then used to create the outer surfaces and produce 3D CAD model of the mirror as shown in Figure 2. The CAD model had 3048 feature parameters.

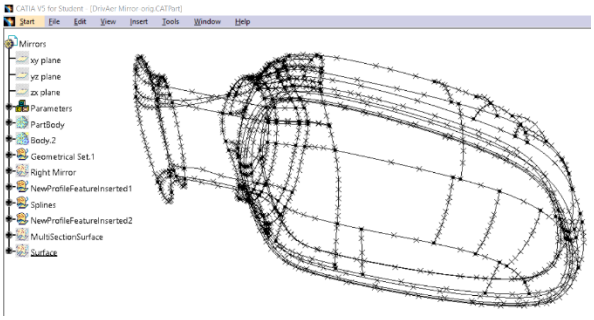


Figure 1: Parametric CAD model of car mirror (wireframe)

In order to capture the CAD surface movement with respect to the change in CAD parameters, the *design velocity* is calculated. This is the movement of the CAD model boundary in the normal direction due to a change in the parameter value and can be mathematically formulated as

$$V_n = \delta X_s \cdot \hat{n}, \tag{1}$$

where δX_s is the movement of surface nodes and $\hat{n}$ is the outward unit normal of the surface of the CAD model at the node location.

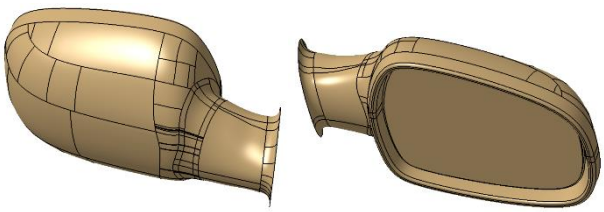


Figure 2: CAD model of car mirror (solid)

The convention adopted throughout this work is that a positive design velocity represents an outward movement of the boundary, and a negative one an inward. Figure 3 shows

design velocity contours produced for the CAD model shown in Figure 2, when some of the CAD parameters defining the model are changed.

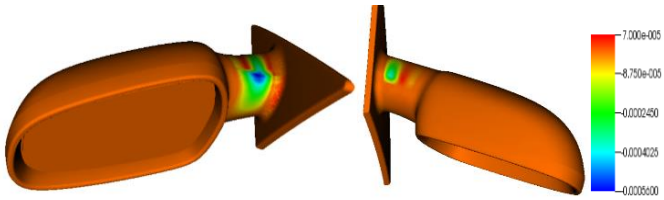


Figure 3: Design velocity for parametric perturbations

The methodology for calculating the design velocities is described by Agarwal *et. al.*[7] and is in effect a finite difference on the shape of the 3D CAD models before and after the parameter perturbation. The implementation process includes calculating the geometrical movement along a normal direction between two discrete representations of the original and perturbed geometry respectively. The process incorporate various CAD design software tools by using a generic representation of CAD model (STEP) format as the CAD format input to the calculation of design velocities. Thus can be easily integrated to most industrial optimisation workflows (regardless of CAD system) and is immune to the topology and labelling issues highlighted by other CAD based optimisation processes[13,14]. It can consider every continuous ("real value") parameter type as an optimisation variable, and it can be adapted to work with any CAD modelling software, as long as it has an API which provides access to the values of the parameters which control the model shape and allows the model geometry to be exported.

3 The continuous adjoint method

The mean-flow problem is governed by the Navier-Stokes equations

$$R^p = \frac{\partial v_i}{\partial x_i} = 0 \tag{2}$$

$$R_i^v = v_j \frac{\partial v_i}{\partial x_j} - \frac{\partial \tau_{ij}}{\partial x_j} + \frac{\partial p}{\partial x_i} = 0, \qquad i = 1,2,3 \tag{3}$$

where p is the static pressure, v_i is the flow velocity, $\tau^{ij} = (\nu + \nu_\tau)\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right)$ are the components of the stress tensor and ν and ν_τ the kinematic and turbulent viscosity respectively.

The turbulence model used is the Spalart-Allmaras turbulence model described by

$$R_i^{\tilde{\nu}} = v_j \frac{\partial \tilde{\nu}}{\partial x_j} - \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\tilde{\nu}}{\sigma}\right)\frac{\partial \tilde{\nu}}{\partial x_j}\right] - \frac{c_{b2}}{\sigma}\left(\frac{\partial \tilde{\nu}}{\partial x_j}\right)^2 - \tilde{\nu}P + \tilde{\nu}D = 0. \tag{4}$$

where P and D are the turbulence production and dissipation terms respectively, and the eddy viscosity coefficient is expressed as

$$\nu_t = \tilde{\nu} f_{v1}(\tilde{\nu}). \tag{5}$$

For the treatment of v_i and ν_t on the wall boundary faces, the wall function technique is employed.

The low frequency noise perceived inside the cabin can be linked to the turbulence level at the area directly outside of the driver side window. In this sense, a surrogate aeroacoustics objective function can be formulated as the integral of the squared turbulent viscosity over a volume near the side window as:

$$F_{noise} = \int_\Omega \nu_t^2 \, d\Omega, \tag{6}$$

The implementation of the continuous adjoint formulation used in this paper to compute the flow sensitivities is the one developed by Papoutsis *et. al.*[10]. For the sake of completeness, a short description of this implementation is also presented here.

The objective function is augmented by the scalar product of the state equations residuals with the adjoint variables, yielding

$$J = F + \int_\Omega u_i R_i^v \, d\Omega + \int_\Omega q R^p \, d\Omega + \int_\Omega \tilde{v}_a R^{\tilde{\nu}} \, d\Omega. \tag{7}$$

The next step is the differentiation of the objective function and the application of the Green-Theorem where necessary. By eliminating the multipliers of terms including the variation of primal variables with respect to the design variables, the adjoint equations, boundary conditions and sensitivity derivatives can be calculated. It is important here to note, that without the differentiation of the turbulence model, relying on the "frozen turbulence" assumption, dealing with an optimization problem of this kind would not be possible, as the objective function itself depends on the turbulent variable $\tilde{\nu}$.

After the aforementioned development the adjoint equations and boundary conditions are derived as

$$R^q = \frac{\partial u_i}{\partial x_i} = 0 \tag{8}$$

$$R_i^u = u_j \frac{\partial v_j}{\partial x_i} - \frac{\partial (v_j u_i)}{\partial x_j} - \frac{\partial \tau_{ij}^a}{\partial x_j} + \frac{\partial q}{\partial x_i} + \tilde{\nu}_\alpha \frac{\partial \tilde{\nu}}{\partial x_i} - \frac{\partial}{\partial x_l}\left(\tilde{\nu}_\alpha \tilde{\nu} \frac{C_Y}{Y} e_{mjk} \frac{\partial v_k}{\partial x_j} e_{mli}\right) = 0, \quad i = 1,2,3 \tag{9}$$

It is interesting to note, that the last term on the adjoint momentum equation stems directly from the differentiation of the turbulence model. Similar to the primal problem, the adjoint Navier-Stokes are coupled with the adjoint turbulence model to give

$$R^{\tilde{\nu}_\alpha} = -\frac{\partial (v_j \tilde{\nu}_\alpha)}{\partial x_j} - \frac{\partial}{\partial x_j}\left[\left(\nu + \frac{\tilde{\nu}}{\sigma}\right)\frac{\partial \tilde{\nu}_\alpha}{\partial x_j}\right] + \frac{1}{\sigma}\frac{\partial \tilde{\nu}_\alpha}{\partial x_j}\frac{\partial \tilde{\nu}}{\partial x_j} + 2\frac{c_{b2}}{\sigma}\frac{\partial}{\partial x_j}\left(\tilde{\nu}_\alpha \frac{\partial \tilde{\nu}}{\partial x_j}\right) + \tilde{\nu}_\alpha \tilde{\nu} C_{\tilde{\nu}} + \frac{\partial \nu_t}{\partial \tilde{\nu}}\frac{\partial u_i}{\partial x_j}\left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i}\right) + (-P + D)\tilde{\nu}_\alpha + \acute{F}_\Omega^{\tilde{\nu}} = 0, \tag{10}$$

A selection of the terms comprising the sensitivity derivatives is given by

$$\frac{\delta J}{\delta b_n} = -\int_S \left[\left(\tau_{ij}^a n_j - q n_i\right) \frac{\partial v_i}{\partial x_k} + u_{\langle t \rangle}^I \frac{\partial \tau_{ij}}{\partial x_k} n_j t_i^I - \nu \frac{\partial \tilde{v}_a}{\partial x_j} n_j \frac{\partial \tilde{v}}{\partial x_k} + T_k \right] \frac{\delta x_k}{\delta b_n} dS \qquad (11)$$

For the complete expression of the sensitivity derivatives, as well as the adjoint boundary conditions, the interested reader may refer to Papoutsis *et. al*[8].

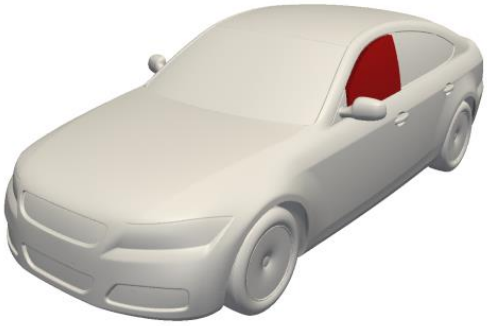


Figure 4: Volume (in red) over which the objective function is integrated. The volume was created by the extrusion of the DrivAer driver window by 3 cm.

Parametric sensitivity is a measure of change in performance (J) caused by the change in the value of a parameter for which a shape change occurs. If the design velocity (V_n) over the surface (A) for a parametric change (dP) is known, along with the adjoint surface sensitivity (φ) the parametric sensitivity, S_i is:

$$\mathrm{S_i} = \frac{\mathrm{d}J}{\mathrm{d}P} = -\int_A \varphi V_n \, dA \qquad (12)$$

4 Effectiveness of CAD parameters

A novel approach to rate the quality of CAD parameters in an optimization step using adjoint sensitivity (S_i) and design velocity (V_n) was presented in Ref[7]. The parametric effectiveness was proposed as the ratio of the change in performance achieved by perturbing all of the parameters in an optimum way subject to the constraint of a given root-mean-squared boundary movement, calculated as[7]

$$\left(\frac{dJ}{dV}\right)_{param} = -\sqrt{\frac{A}{\int_A \left(\sum_{i=1}^n S_i V_{ni}\right)^2 dA}} \sum_{i=1}^n (S_i)^2 , \qquad (13)$$

to the change in performance achieved by allowing the boundary of the model to move freely, calculated as[7]

$$\left(\frac{dJ}{dV}\right)_{max} = -\sqrt{A \int_{A_j} \varphi^2 dA} . \qquad (14)$$

A low parametric effectiveness indicates that the parameters in the model will not be able to perturb the model shape in the manner the adjoint sensitivity map or, in doing so, their change will also cause large movements of the body in less sensitive regions. It indicates it may be better to reparametrize the model before attempting optimisation.

As stated before, a complex CAD model could be defined using hundreds or thousands of CAD feature parameters. While all could be used for optimization, doing so would not be the most efficient method of optimizing a component[7]. In addition to that, using a large parameter set giving a larger boundary movement may also have an impact on the optimization of other objective functions.

In this paper, parametric effectiveness combined with a power-set approach is used to ascertain the most effective parametric combination to be used for optimization. In mathematics, the power-set of any set $\mathbb{Q}$ of n parameters is the set of all subsets of $\mathbb{Q}$ (including the empty set) giving a total of $(2^n - 1)$ different parametric combinations.

The methodology to advice this is firstly, to develop a link between CAD modelling software CATIA V5 and adjoint sensitivities to calculate the effectiveness of the CAD model parameters. This is used as the first criteria to downselect the set of parameters. Secondly, the power-set of the selected parameter set is formulated to get different combinations of these CAD parameters. Thirdly, the parametric effectiveness of these parameter combinations is calculated to select the parameters to get the parametric combination with highest parametric effectiveness.

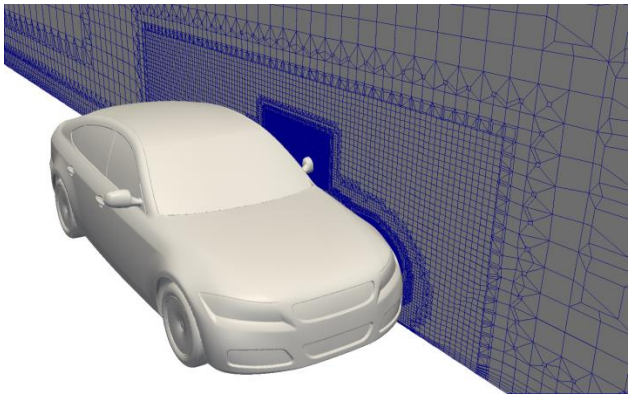


Figure 5: Slice of the computational grid around the drivAer vehicle. The refinement boxes around the car and the mirror can be seen.

5 Results

In this paper, the developed approach is demonstrated for the aerodynamic shape optimization of an automotive mirror, targeting at minimization the surrogate objective function for aeroacoustics, presented in section 3. For the flow and adjoint analysis, half of the car was meshed. The computational grid consisted of 5 million cells. As seen in Figure 5, a refinement box was defined around the mirror for a better turbulence resolution in this area.

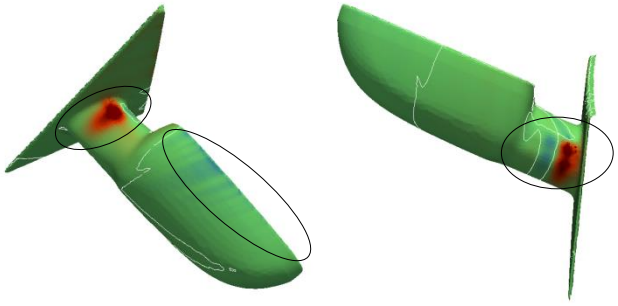


Figure 6: Sensitivity maps targetting at turbulent noise minimisation, seen from top (left) and bottom (right).

The adjoint sensitivity map for minimizing noise inside the car is computed at the first optimization cycle and is presented in Figure 6, where red areas must be pushed inward to the surface while blue are to be pulled outward in order to reduce the objective function.

For the CAD model shown in Figure 1, a total of 3048 CAD parameters were defined which controlled the position of individual points. The CPU time required to update 3048 CAD parameters is nearly 3 hours, and makes the optimization to be computationally expensive. Thus, the set of parameters were firstly reduced to a total of 339, by selecting those which moved the boundary in the regions shown in Figure 6. These were identified manually based on selecting the features in the areas of high adjoint sensitivities, and also on the geometrical constraints of the model. An overall parametric effectiveness of these 339 parameters was calculated as 0.61. Since, the calculation of parametric effectiveness assumes that the boundary movement caused by changing individual parameters are linearly independent[8], it may result in a number of redundant parameters moving the similar areas of the boundary. This leads to a reduction in overall parametric effectiveness when the parametric combinations are analysed.

The computational requirement for evaluating the parametric effectiveness of all possible subsets as given by power-set is very high, so a small set of parameters are analysed in this work. The set of 10 CAD parameters with highest parametric effectiveness were selected giving an overall parametric effectivenss of 0.58. Further, the parametric effectiveness of all possible combinations of these 10 CAD parameters (as given by power-set) was analysed. For a combination of 7 different CAD parameters, an overall parametric effectiveness of 0.61 was achieved. Note that these parameters were not the ones with highest parametric effectiveness (The combination of 7 parameters with highest parametric effectiveness resulted in an overall parametric effectiveness of 0.50).

The shape optimization of car mirror was performed using the 7 CAD parameters resulted from power-set approach, and a steepest descent strategy to update the design variables with the computed gradients. After the optimization algorithm converged, the optimal geometry was by 5.14% “quieter”. Comparing the starting and optimized geometries (Figure 7), it is seen that the top of the neck of the mirror has been pushed in to suppress the generation of turbulence on the wake of the mirror and consequently to minimize the turbulence viscosity flowing through the volume over which the objective function is integrated (Figure 8).

6 Conclusions

In this paper, a CAD-based shape optimization driven by the adjoint sensitivities for a subset of parameters was presented. The CAD parameters were selected based on a power-set approach, which allowed the selection of parameters in the regions of high adjoint sensitivities to be used in optimization. The method was applied to the drivAer vehicle, targeting at turbulent noise reduction over a volume residing next to the driver window. The optimization process altered the CAD-based geometry of the mirror and successfully reduced the objective function.

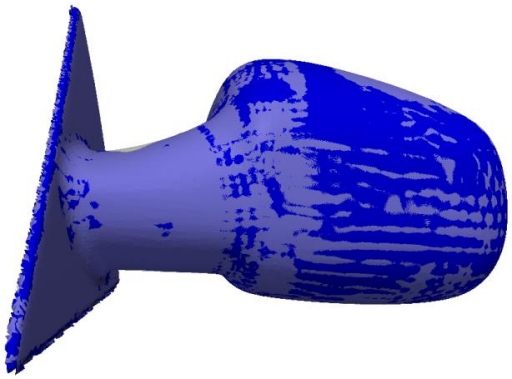


Figure 7: Comparison between the starting (transparent) and blue optimised (blue) geometry. It is seen, that accordingly to the adjoint sensitivities the neck of the mirror is squeezed, in order to minimise the turbulent noise.

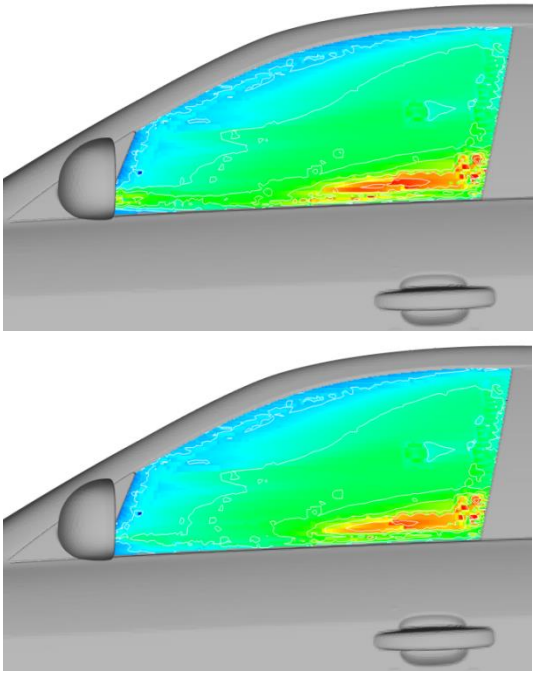


Figure 8: Squared turbulent viscocity computed at a slice of the volume over which the objective function is integrated. The turbulent viscosity is reduced at the wake of the mirror neck, thanks to its optimised shape (starting geometry top, optimised bottom).

7 Acknowledgements

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